

LONESOME PINE, VA, USA

WMO: 724117

Lat: 36.988N

Lon: 82.530W

Elev: 818

StdP: 91.88

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 63802

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.9	-11.0	-19.0	0.8	-7.1	-17.1	0.9	-7.1	11.1	6.5	10.1	5.9	2.7	300	0.427

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.3	30.2	20.8	28.9	20.6	27.7	20.1	23.2	26.9	22.5	26.1	22.0	25.5	2.6	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.4	18.9	24.6	22.0	18.4	24.2	21.1	17.4	23.5	73.4	26.8	71.0	26.2	68.5	26.1	27.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.0	7.9	7.0	-16.9	32.3	3.8	1.6	-19.6	33.5	-21.8	34.5	-23.9	35.4	-26.7	36.6		
(5)			-17.1	24.3	2.9	1.1	-19.2	25.0	-20.9	25.7	-22.5	26.3	-24.6	27.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	12.5	0.8	2.9	7.2	12.7	17.0	20.7	22.6	22.2	19.0	13.3	7.5	3.3	(6)
(7)		DBStd	9.00	6.73	6.39	6.10	5.21	4.09	2.83	2.16	2.14	3.58	4.83	5.34	5.84	(7)
(8)		HDD10.0	1014	292	209	126	33	4	0	0	0	1	24	108	218	(8)
(9)		HDD18.3	2571	542	432	346	176	72	12	2	1	34	164	324	466	(9)
(10)		CDD10.0	1926	8	11	40	114	222	320	389	378	272	127	34	11	(10)
(11)		CDD18.3	441	0	0	1	8	32	82	133	121	55	9	1	0	(11)
(12)		CDH23.3	2573	0	0	4	42	176	494	788	684	336	48	1	0	(12)
(13)		CDH26.7	536	0	0	0	3	17	104	181	153	71	6	0	0	(13)
(14)	Wind	WSAvg	2.7	3.5	3.5	3.4	3.5	2.7	2.2	2.0	1.8	2.0	2.5	2.9	3.2	(14)
(15)	Precipitation	PrecAvg	1229	114	104	124	104	111	110	129	101	76	59	82	115	(15)
(16)		PrecMax	1504	172	252	259	229	190	176	202	180	179	112	154	242	(16)
(17)		PrecMin	918	57	44	52	47	37	64	49	43	21	7	28	43	(17)
(18)		PrecStd	173	32	48	50	47	44	33	41	32	44	26	34	49	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.1	20.0	23.6	27.1	28.0	31.2	31.9	31.3	30.9	27.3	22.0	17.8	(19)
(20)			MCWB	12.8	13.2	14.1	16.0	19.2	21.5	21.9	21.3	18.6	18.6	13.9	13.3	(20)
(21)		2%	DB	14.9	17.0	21.0	24.1	27.1	29.0	30.1	29.8	28.6	24.2	18.9	16.1	(21)
(22)			MCWB	11.5	12.2	12.7	14.7	18.2	20.7	21.4	21.0	18.8	16.7	12.2	13.2	(22)
(23)		5%	DB	12.8	14.0	17.9	22.6	25.6	27.7	28.6	28.0	27.3	22.5	17.4	13.7	(23)
(24)			MCWB	10.0	9.8	11.2	13.8	17.9	19.9	21.1	20.7	18.7	15.9	11.9	10.8	(24)
(25)		10%	DB	11.1	12.2	16.2	21.1	23.7	26.3	27.4	27.3	25.2	20.9	15.9	12.3	(25)
(26)			MCWB	8.8	8.6	10.4	13.5	17.2	19.5	20.7	20.6	18.1	15.6	11.4	9.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.1	15.8	15.6	18.0	22.2	23.9	24.2	24.1	22.8	21.3	16.3	15.9	(27)
(28)			MCD B	15.6	17.3	20.5	23.1	25.8	28.0	28.2	27.6	26.1	24.3	17.9	16.6	(28)
(29)		2%	WB	12.5	13.3	14.2	16.9	20.5	22.5	23.2	23.0	21.7	19.6	14.9	13.7	(29)
(30)			MCD B	14.3	15.6	18.5	21.8	24.3	26.6	26.9	26.6	25.0	22.8	17.4	15.2	(30)
(31)		5%	WB	10.9	11.2	12.9	15.8	19.5	21.5	22.6	22.3	20.7	17.7	13.4	12.2	(31)
(32)			MCD B	12.7	13.1	16.5	20.1	23.5	25.3	26.2	25.9	23.6	20.0	16.4	13.2	(32)
(33)		10%	WB	8.7	9.1	11.6	14.7	18.4	20.7	22.0	21.6	19.6	16.3	12.0	10.0	(33)
(34)			MCD B	10.4	11.4	15.2	19.0	21.9	24.5	25.4	25.0	23.1	19.2	14.4	11.9	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.5	8.7	9.9	10.6	10.2	10.0	9.3	9.5	9.9	9.9	9.3	8.1	(35)
(36)			MCDBR	9.1	10.5	11.5	11.9	11.3	11.3	10.8	10.6	11.7	11.5	10.4	8.8	(36)
(37)			MCWBR	6.9	7.3	6.5	5.8	5.2	5.1	4.4	4.5	4.5	5.4	6.3	6.6	(37)
(38)		5% WB	MCDBR	7.9	8.8	9.6	9.8	9.8	9.8	9.0	8.9	9.0	9.2	8.5	7.7	(38)
(39)			MCWBR	6.9	7.7	6.5	6.0	5.3	5.3	4.5	4.6	4.9	5.7	6.5	7.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.285	0.296	0.318	0.355	0.392	0.417	0.446	0.450	0.364	0.334	0.302	0.282	(40)	
(41)		taud	2.553	2.515	2.491	2.391	2.283	2.242	2.193	2.161	2.396	2.486	2.554	2.581	(41)	
(42)		Ebn at Noon	919	947	950	925	891	865	836	824	888	888	889	900	(42)	
(43)		Edh at Noon	76	89	100	117	133	139	144	146	109	91	75	69	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.08	2.67	3.86	5.02	5.60	6.10	5.69	5.38	4.57	3.48	2.53	1.84	(44)	
(45)		RadStd	0.19	0.36	0.33	0.36	0.44	0.36	0.35	0.26	0.40	0.43	0.28	0.19	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.62	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (30 neighbors)		+0.46	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+137	+79

Nomenclature: See separate page